

Steepest Descent

Learning objectives

By the end of this chapter, you should be able to:

- obtain the negative-gradient direction from an isotropic quadratic model;
- explain why this direction is the Euclidean direction of steepest local decrease;
- apply steepest descent with exact line search or Armijo backtracking;
- use smoothness and the descent lemma to certify sufficient decrease;
- explain zig-zagging and the role of conditioning on quadratic objectives;
- prove a linear function-value rate under strong convexity and smoothness;
- derive preconditioned steepest descent and its effective condition number;
- distinguish deterministic steepest descent from stochastic gradient descent.

Chapter 9 started from a second-order model. Pure Newton uses the Hessian directly. Shifted Newton replaces it by $\mathbf{H}_f(\mathbf{x}^{(k)}) + \tau^{(k)} \mathbf{I} \succ 0$, whereas Newton–Armijo only needs a safeguard that produces a descent direction before backtracking; that safeguard does not itself require a positive-definite shifted matrix. A typical line-search method first chooses a descent direction and then selects a step length by exact minimization or by a sufficient-decrease rule such as Armijo.

This chapter first keeps the same framework but removes the Hessian. The resulting direction is inexpensive, always descends at a nonstationary point, and is the fastest local descent direction in the Euclidean norm. Its main weakness—slow progress across a narrow valley—leads naturally to the condition number and then to preconditioning, which replaces Euclidean geometry by a carefully chosen inexpensive metric.

Throughout this chapter, $\mathcal{D} = \mathbb{R}^n$ and $f : \mathcal{D} \rightarrow \mathbb{R}$, unless another domain is stated explicitly.

10.1 From the Newton model to the negative gradient

At $\mathbf{x}^{(k)}$, write

$$\mathbf{g}^{(k)} := \nabla f(\mathbf{x}^{(k)}).$$

Replace the Hessian in the second-order model by the identity matrix:

$$m^{(k)}(\mathbf{d}) := f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^\top \mathbf{d} + \frac{1}{2} \|\mathbf{d}\|^2. \quad (10.1)$$

This model is strictly convex because its Hessian with respect to $\mathbf{d}$ is $\mathbf{I}$. Completing the square gives

$$m^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) - \frac{1}{2} \|\mathbf{g}^{(k)}\|^2 + \frac{1}{2} \|\mathbf{d} + \mathbf{g}^{(k)}\|^2,$$

so its unique minimizer is

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}.$$

Replacing $\mathbf{I}$ by $c\mathbf{I}$, where $c > 0$, only changes the minimizer to $-\mathbf{g}^{(k)}/c$, which lies on the same search ray. Exact line search gives the same next iterate after the step length is reparameterized. For Armijo backtracking, the same conclusion requires the initial trial step, and hence the backtracking grid, to be rescaled by c ; keeping the same numerical grid can produce a different accepted point. Thus $\mathbf{I}$ is sufficient for deriving the search ray, but the scaling convention still matters to an implemented inexact line search.

10.1.1 Why the direction is steepest

Let $\mathbf{x} \in \mathcal{D}$ and suppose that f is differentiable at $\mathbf{x}$. For a unit vector $\mathbf{u}$,

$$D_{\mathbf{u}}f(\mathbf{x}) = \nabla f(\mathbf{x})^T \mathbf{u}$$

is the instantaneous rate of change along $\mathbf{u}$.

Proposition 10.1 (Euclidean steepest descent direction). *If $\nabla f(\mathbf{x}) \neq \mathbf{0}$, then*

$$\min_{\|\mathbf{u}\|=1} D_{\mathbf{u}}f(\mathbf{x}) = -\|\nabla f(\mathbf{x})\|,$$

and the unique minimizing direction is

$$\mathbf{u}_* = -\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}.$$

Consequently, $-\nabla f(\mathbf{x})$ is a strict descent direction.

Proof. For every unit vector $\mathbf{u}$, Cauchy–Schwarz gives

$$\nabla f(\mathbf{x})^T \mathbf{u} \geq -\|\nabla f(\mathbf{x})\| \|\mathbf{u}\| = -\|\nabla f(\mathbf{x})\|.$$

Equality in Cauchy–Schwarz requires $\mathbf{u}$ to be a negative multiple of $\nabla f(\mathbf{x})$. The unit-length requirement therefore gives the unique choice $\mathbf{u}_* = -\nabla f(\mathbf{x})/\|\nabla f(\mathbf{x})\|$. Finally,

$$\nabla f(\mathbf{x})^T [-\nabla f(\mathbf{x})] = -\|\nabla f(\mathbf{x})\|^2 < 0,$$

which proves the descent claim. □

The directional-derivative statement also gives actual decrease, rather than only an infinitesimal interpretation. Differentiability implies

$$f(\mathbf{x} - \alpha \nabla f(\mathbf{x})) = f(\mathbf{x}) - \alpha \|\nabla f(\mathbf{x})\|^2 + o(\alpha).$$

If $\nabla f(\mathbf{x}) \neq \mathbf{0}$, then for all sufficiently small $\alpha > 0$, the remainder has magnitude at most $\alpha \|\nabla f(\mathbf{x})\|^2/2$. Hence

$$f(\mathbf{x} - \alpha \nabla f(\mathbf{x})) \leq f(\mathbf{x}) - \frac{\alpha}{2} \|\nabla f(\mathbf{x})\|^2 < f(\mathbf{x}).$$

Warning

The word “steepest” is meaningless without a normalization. Multiplying a direction by an arbitrarily large positive number makes its directional derivative arbitrarily negative. Moreover, the negative gradient is steepest for the Euclidean norm; another norm generally gives another direction.

10.2 Steepest descent with line search

If $\mathbf{g}^{(k)} \neq \mathbf{0}$, the choice $\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$ satisfies $(\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)} < 0$. We may therefore choose the step length by either of the two line-search rules reviewed in [Chapter 9](#).

Exact line search, when its minimum is attained, chooses

$$\alpha^{(k)} \in \arg \min_{\alpha \geq 0} f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}). \quad (10.2)$$

For Armijo backtracking, fix $\sigma, \beta \in (0, 1)$ and an initial trial step $\bar{\alpha} > 0$. Test $\bar{\alpha}, \beta\bar{\alpha}, \beta^2\bar{\alpha}, \dots$ until

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + \sigma \alpha (\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)}. \quad (10.3)$$

For $\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$, the last term is $-\sigma \alpha \|\mathbf{g}^{(k)}\|^2$.

Algorithm: Line-search steepest descent

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, a gradient tolerance $\varepsilon_g > 0$, a maximum number K of updates, and either exact line search or Armijo backtracking. For $k = 0, \dots, K - 1$:

1. Compute $\mathbf{g}^{(k)} := \nabla f(\mathbf{x}^{(k)})$.
2. If $\|\mathbf{g}^{(k)}\| \leq \varepsilon_g$, stop and return $\mathbf{x}^{(k)}$.
3. Set $\mathbf{d}^{(k)} := -\mathbf{g}^{(k)}$.
4. Choose $\alpha^{(k)} > 0$ by [Equation \(10.2\)](#), or use Armijo backtracking until [Equation \(10.3\)](#) holds.
5. Set $\mathbf{x}^{(k+1)} := \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}$.

After the last permitted update, compute $\mathbf{g}^{(K)}$ and apply the stopping test once more. Otherwise return $\mathbf{x}^{(K)}$ with the status that the maximum number of updates was reached.

Exact line search asks for the best point on the current ray. Armijo backtracking only asks for enough decrease, and is usually cheaper. Neither choice changes the direction $-\mathbf{g}^{(k)}$.

10.3 Smoothness, descent, and majorization

Definition 10.1 (Smoothness). Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open and convex. A differentiable function $f : \mathcal{D} \rightarrow \mathbb{R}$ is L -smooth on $\mathcal{D}$, where $L > 0$, if

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\| \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathcal{D}.$$

For a base point $\mathbf{x} \in \mathcal{D}$ and a displacement $\mathbf{d}$ with $\mathbf{x} + \mathbf{d} \in \mathcal{D}$, define the quadratic upper model

$$M_{L,\mathbf{x}}(\mathbf{d}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{d} + \frac{L}{2} \|\mathbf{d}\|^2. \quad (10.4)$$

Lemma 10.2 (Descent lemma). *If f is L -smooth on an open convex set $\mathcal{D}$, then*

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2 \quad (\mathbf{x}, \mathbf{y} \in \mathcal{D}). \quad (10.5)$$

Proof. Fix $\mathbf{x}, \mathbf{y} \in \mathcal{D}$, put $\mathbf{d} := \mathbf{y} - \mathbf{x}$, and define $\phi(t) := f(\mathbf{x} + t\mathbf{d})$ for $t \in [0, 1]$. Convexity of $\mathcal{D}$ keeps this segment in the domain. The fundamental theorem of calculus gives

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{d} &= \int_0^1 [\nabla f(\mathbf{x} + t\mathbf{d}) - \nabla f(\mathbf{x})]^\top \mathbf{d} dt \\ &\leq \int_0^1 \|\nabla f(\mathbf{x} + t\mathbf{d}) - \nabla f(\mathbf{x})\| \|\mathbf{d}\| dt \\ &\leq \int_0^1 Lt \|\mathbf{d}\|^2 dt = \frac{L}{2} \|\mathbf{d}\|^2. \end{aligned}$$

This is Equation (10.5). □

The upper model need not be the objective's Taylor polynomial. For example, consider the visibly nonquadratic one-dimensional function

$$\phi(t) := \frac{1}{2}t^2 + \frac{1}{5}\sin(2t). \quad (10.6)$$

Since

$$\phi''(t) = 1 - \frac{4}{5}\sin(2t) \in \left[\frac{1}{5}, \frac{9}{5}\right],$$

the function is L -smooth with $L = 9/5$. At every base point $\bar{t}$, the graph of $\phi(\bar{t} + d)$ lies below $M_{L,\bar{t}}(d)$, as illustrated in Figure 10.1.

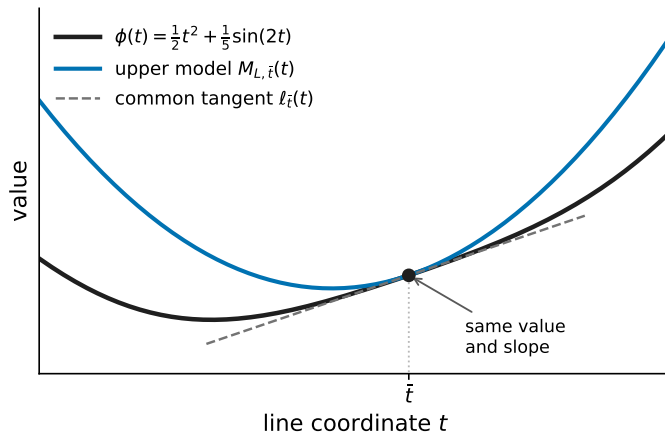


Figure 10.1: The quadratic upper model for the nonquadratic function in Equation (10.6). The model agrees with the function in value and slope at the base point and majorizes it globally.

Corollary 10.3 (Finite termination of Armijo backtracking). *Let f be L -smooth, let $\mathbf{g} = \nabla f(\mathbf{x})$, and let $\mathbf{d}$ satisfy $\mathbf{g}^\top \mathbf{d} < 0$. Fix $\sigma, \beta \in (0, 1)$ and $\bar{\alpha} > 0$. Every trial step satisfying*

$$0 < \alpha \leq \frac{2(1-\sigma)(-\mathbf{g}^\top \mathbf{d})}{L \|\mathbf{d}\|^2} \quad (10.7)$$

obeys the Armijo inequality, provided the segment from $\mathbf{x}$ to $\mathbf{x} + \alpha \mathbf{d}$ lies in $\mathcal{D}$. Consequently, if all sufficiently short trial segments remain in $\mathcal{D}$, Armijo backtracking terminates after finitely many reductions.

Proof. The descent lemma gives

$$f(\mathbf{x} + \alpha \mathbf{d}) \leq f(\mathbf{x}) + \alpha \mathbf{g}^\top \mathbf{d} + \frac{L\alpha^2}{2} \|\mathbf{d}\|^2.$$

The bound in Equation (10.7) is equivalent to

$$\frac{L\alpha}{2} \|\mathbf{d}\|^2 \leq (1-\sigma)(-\mathbf{g}^\top \mathbf{d}).$$

Substitution yields

$$f(\mathbf{x} + \alpha \mathbf{d}) \leq f(\mathbf{x}) + \sigma \alpha \mathbf{g}^\top \mathbf{d},$$

which is the Armijo condition. Finally, $\beta^j \bar{\alpha} \rightarrow 0$ as $j \rightarrow \infty$, so the tested sequence eventually falls below the positive threshold in Equation (10.7). $\square$

Suppose $\mathbf{x} \in \mathcal{D}$ and the trial point $\mathbf{x} - \alpha \nabla f(\mathbf{x})$ also belongs to $\mathcal{D}$. Substituting this trial point into the descent lemma yields

$$f(\mathbf{x} - \alpha \nabla f(\mathbf{x})) \leq f(\mathbf{x}) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(\mathbf{x})\|^2. \quad (10.8)$$

Thus every $0 < \alpha \leq 1/L$ satisfies

$$f(\mathbf{x} - \alpha \nabla f(\mathbf{x})) \leq f(\mathbf{x}) - \frac{\alpha}{2} \|\nabla f(\mathbf{x})\|^2, \quad (10.9)$$

which is exactly the Armijo condition with $\sigma = 1/2$. The right-hand side of Equation (10.8), viewed as a quadratic upper bound in α , is minimized at $\alpha = 1/L$. In particular,

$$f\left(\mathbf{x} - \frac{1}{L} \nabla f(\mathbf{x})\right) \leq f(\mathbf{x}) - \frac{1}{2L} \|\nabla f(\mathbf{x})\|^2,$$

which we call the *descent inequality*.

The identity-model direction and the global upper model play different roles: the first chooses a ray, whereas the second certifies a step length. For $\alpha > 0$, define

$$q_\alpha^{(k)}(\mathbf{y}) := f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^\top (\mathbf{y} - \mathbf{x}^{(k)}) + \frac{1}{2\alpha} \|\mathbf{y} - \mathbf{x}^{(k)}\|^2. \quad (10.10)$$

Then

$$q_\alpha^{(k)}(\mathbf{y}) = M_{1/\alpha, \mathbf{x}^{(k)}}(\mathbf{y} - \mathbf{x}^{(k)}).$$

Proposition 10.4 (Quadratic majorization and the gradient step). *Suppose f is L -*

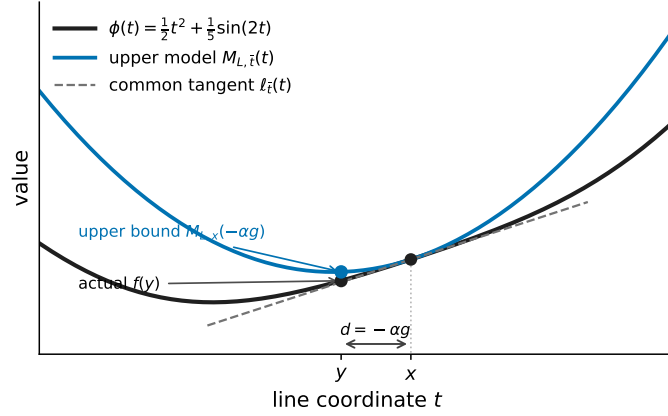


Figure 10.2: The gradient trial point $\mathbf{y} = \mathbf{x} - \alpha \nabla f(\mathbf{x})$ is evaluated through the upper model: $f(\mathbf{y}) \leq M_{L,\mathbf{x}}(-\alpha \nabla f(\mathbf{x}))$. The plotted example uses the nonquadratic function in Equation (10.6).

smooth on an open convex set $\mathcal{D}$, let $\mathbf{x}^{(k)} \in \mathcal{D}$, and let $0 < \alpha \leq 1/L$. Then

$$f(\mathbf{y}) \leq q_{\alpha}^{(k)}(\mathbf{y}) \quad \text{for every } \mathbf{y} \in \mathcal{D},$$

with equality at $\mathbf{y} = \mathbf{x}^{(k)}$. The unique unconstrained minimizer of $q_{\alpha}^{(k)}$ is $\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}$. If this trial point belongs to $\mathcal{D}$, it satisfies Equation (10.9).

Proof. Because $1/\alpha \geq L$, the descent lemma gives

$$\begin{aligned} f(\mathbf{y}) &\leq f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^{\top}(\mathbf{y} - \mathbf{x}^{(k)}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}^{(k)}\|^2 \\ &\leq q_{\alpha}^{(k)}(\mathbf{y}). \end{aligned}$$

Both inequalities are equalities at $\mathbf{y} = \mathbf{x}^{(k)}$. Completing the square in Equation (10.10) shows that its unique unconstrained minimizer is $\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}$. If this point belongs to $\mathcal{D}$, then

$$q_{\alpha}^{(k)}(\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}) = f(\mathbf{x}^{(k)}) - \frac{\alpha}{2} \|\mathbf{g}^{(k)}\|^2.$$

Majorization proves the claimed decrease. $\square$

Warning

The bound $\alpha \leq 1/L$ is sufficient, not necessary, for a particular step to satisfy Armijo. For example, let $f(x_1, x_2) = \frac{1}{2}(x_1^2 + 10x_2^2)$. Its tight smoothness constant is $L = 10$, but at $\mathbf{x} = (1, 0)$, the step $\alpha = 1/2 > 1/L$ along the negative gradient satisfies Armijo with $\sigma = 1/2$. The global constant must cover every direction, whereas this step only sees curvature in the x_1 -direction.

10.4 Exact line search on strictly convex quadratics

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be symmetric positive definite and let $\mathbf{c} \in \mathbb{R}^n$. On $\mathcal{D} = \mathbb{R}^n$, consider

$$q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^{\top} \mathbf{Q} \mathbf{x} - \mathbf{c}^{\top} \mathbf{x}. \quad (10.11)$$

Its unique minimizer is $\mathbf{x}^* = \mathbf{Q}^{-1}\mathbf{c}$, and $\mathbf{g}^{(k)} = \mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}$.

Proposition 10.5 (Exact steepest-descent step for a quadratic). *If $\mathbf{g}^{(k)} \neq \mathbf{0}$, exact line search along $\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$ gives*

$$\alpha^{(k)} = \frac{(\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)}}{(\mathbf{g}^{(k)})^\top \mathbf{Q} \mathbf{g}^{(k)}}. \quad (10.12)$$

Proof. Expanding the line objective gives

$$q(\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}) = q(\mathbf{x}^{(k)}) - \alpha (\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)} + \frac{\alpha^2}{2} (\mathbf{g}^{(k)})^\top \mathbf{Q} \mathbf{g}^{(k)}.$$

Its second derivative is $(\mathbf{g}^{(k)})^\top \mathbf{Q} \mathbf{g}^{(k)} > 0$, so the line objective is strictly convex. Setting its first derivative to zero gives Equation (10.12); the value is positive and is therefore also the unique minimizer over $\alpha \geq 0$. $\square$

Example 10.1 (A rotated quadratic family with controlled shape). Let

$$q_r(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top \mathbf{Q}_r \mathbf{x}, \quad \mathbf{Q}_r := \mathbf{U} \begin{pmatrix} 1 & 0 \\ 0 & r \end{pmatrix} \mathbf{U}^\top, \quad r \geq 1, \quad (10.13)$$

where

$$\mathbf{U} := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

The eigenvalues of $\mathbf{Q}_r$ are 1 and r , so r is the ratio of its largest to its smallest curvature. For every member of the family, use the same initialization rule in eigen-coordinates,

$$\mathbf{U}^\top \mathbf{x}^{(0)} = (4, 4/r)^\top. \quad (10.14)$$

With exact line search, $r = 5$ requires 18 updates to reduce the relative objective gap below 10^{-6} , whereas $r = 50$ requires 173 updates; see Figure 10.3 for an illustration. Both runs optimize every search line exactly. The difference comes from the more elongated landscape, which makes the iterates cross the valley many more times. The exact count follows from the recurrence derived in Equation (10.22).

Theorem 10.6 (Orthogonality under exact line search). *Suppose an exact line minimizer in Equation (10.2) is attained and $\mathbf{g}^{(k)} \neq \mathbf{0}$. Then*

$$(\mathbf{g}^{(k+1)})^\top \mathbf{g}^{(k)} = 0.$$

If the next update is also made, the two successive displacement vectors $\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}$ and $\mathbf{x}^{(k+2)} - \mathbf{x}^{(k+1)}$ are orthogonal.

Proof. Define

$$h(\alpha) := f(\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}), \quad \alpha \geq 0.$$

Because $h'(0) = -\|\mathbf{g}^{(k)}\|^2 < 0$, an attained minimizer cannot be 0. It is therefore an interior point $\alpha^{(k)} > 0$, at which

$$0 = h'(\alpha^{(k)}) = -\nabla f(\mathbf{x}^{(k)} - \alpha^{(k)} \mathbf{g}^{(k)})^\top \mathbf{g}^{(k)} = -(\mathbf{g}^{(k+1)})^\top \mathbf{g}^{(k)}.$$

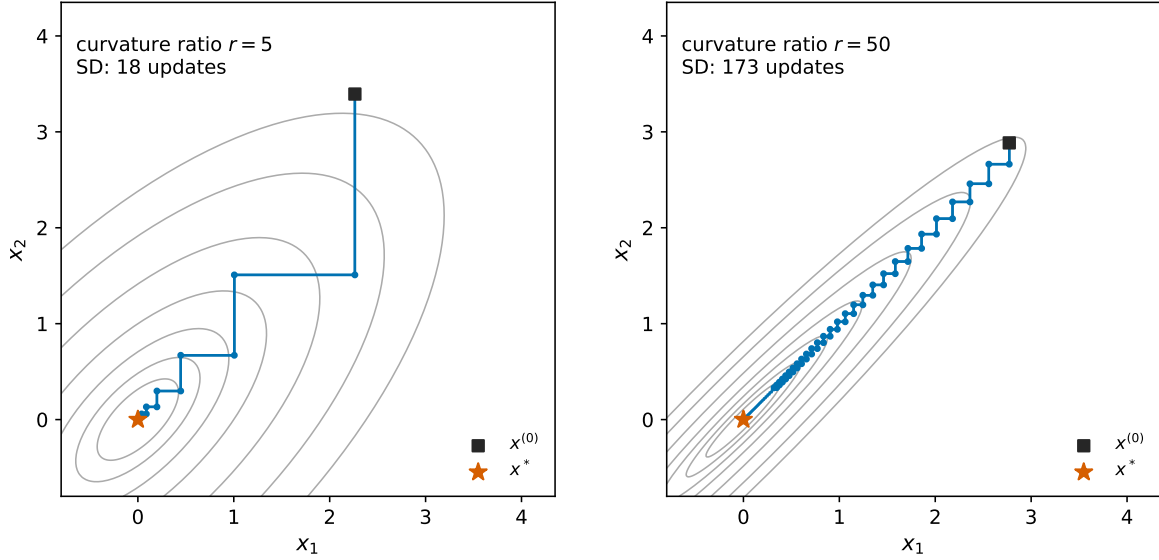


Figure 10.3: Exact-line-search steepest descent on the rotated quadratic family for curvature ratios $r = 5$ (left) and $r = 50$ (right).

Finally,

$$\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)} = -\alpha^{(k)} \mathbf{g}^{(k)}, \quad \mathbf{x}^{(k+2)} - \mathbf{x}^{(k+1)} = -\alpha^{(k+1)} \mathbf{g}^{(k+1)},$$

so their inner product is zero whenever both displacements are defined. $\square$

The orthogonality is a property of exact line search, not a guarantee of fast convergence. It explains why an elongated landscape can force the method to alternate between nearly opposing sides of a valley.

Newton's method provides the useful contrast. For the general quadratic in Equation (10.11),

$$\mathbf{d}_N^{(k)} = -\mathbf{Q}^{-1} \mathbf{g}^{(k)} = -\mathbf{Q}^{-1} (\mathbf{Q} \mathbf{x}^{(k)} - \mathbf{c}) = \mathbf{x}^* - \mathbf{x}^{(k)}.$$

Thus a full Newton step reaches $\mathbf{x}^*$ in one update from every starting point. For a quadratic, the second-order model is the objective itself.

10.5 Strong convexity, smoothness, and a linear rate

Definition 10.2 (Strong convexity). Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open and convex and let $\mu > 0$. A differentiable function $f : \mathcal{D} \rightarrow \mathbb{R}$ is μ -strongly convex on $\mathcal{D}$ if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2 \quad (\mathbf{x}, \mathbf{y} \in \mathcal{D}). \quad (10.15)$$

Proposition 10.7 (Equivalent characterizations of strong convexity). Under the assumptions of the definition, suppose in addition that $f \in C^1(\mathcal{D})$. Then the following statements are equivalent:

- (i) f satisfies Equation (10.15);
- (ii) $h(\mathbf{x}) := f(\mathbf{x}) - \mu \|\mathbf{x}\|^2 / 2$ is convex;

(iii) for all $\mathbf{x}, \mathbf{y} \in \mathcal{D}$ and $\lambda \in [0, 1]$,

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) - \frac{\mu}{2} \lambda (1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2. \quad (10.16)$$

Proof. The gradient of h is $\nabla h(\mathbf{x}) = \nabla f(\mathbf{x}) - \mu \mathbf{x}$. By the tangent-plane characterization in Theorem 4.4, h is convex if and only if

$$f(\mathbf{y}) - \frac{\mu}{2} \|\mathbf{y}\|^2 \geq f(\mathbf{x}) - \frac{\mu}{2} \|\mathbf{x}\|^2 + (\nabla f(\mathbf{x}) - \mu \mathbf{x})^\top (\mathbf{y} - \mathbf{x}).$$

Using

$$\|\mathbf{y}\|^2 - \|\mathbf{x}\|^2 - 2\mathbf{x}^\top (\mathbf{y} - \mathbf{x}) = \|\mathbf{y} - \mathbf{x}\|^2$$

shows that this inequality is exactly Equation (10.15). Hence (i) and (ii) are equivalent.

Next, convexity of h is equivalent to

$$h(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda h(\mathbf{x}) + (1 - \lambda) h(\mathbf{y}).$$

Substitute the definition of h and use

$$\|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\|^2 = \lambda \|\mathbf{x}\|^2 + (1 - \lambda) \|\mathbf{y}\|^2 - \lambda(1 - \lambda) \|\mathbf{x} - \mathbf{y}\|^2.$$

The result is Equation (10.16), proving the equivalence of (ii) and (iii). $\square$

The chord inequality immediately gives

$$\text{strongly convex} \implies \text{strictly convex} \implies \text{convex}.$$

Indeed, when $\mathbf{x} \neq \mathbf{y}$, $\lambda \in (0, 1)$, and $\mu > 0$, the last term in Equation (10.16) is strictly negative.

Warning

Neither converse holds. On $\mathbb{R}$, $t \mapsto t^2$ is strongly convex; $t \mapsto t^4$ is strictly convex but not strongly convex, because strong convexity at the base point 0 would require $t^4 \geq \mu t^2/2$ for every t ; and $t \mapsto 0$ is convex but not strictly convex. Only strong convexity supplies a positive uniform quadratic lower bound.

Proposition 10.8 (Existence and uniqueness on $\mathbb{R}^n$). *Every differentiable strongly convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has a unique global minimizer.*

Proof. Fix $\mathbf{x}_0 \in \mathbb{R}^n$. Strong convexity gives

$$f(\mathbf{y}) \geq f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top (\mathbf{y} - \mathbf{x}_0) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}_0\|^2.$$

The right-hand side tends to $+\infty$ as $\|\mathbf{y}\| \rightarrow \infty$, so f is coercive. Since differentiability implies continuity, the nonempty sublevel set

$$S := \{\mathbf{y} : f(\mathbf{y}) \leq f(\mathbf{x}_0)\}$$

is closed; coercivity makes it bounded. It is therefore compact, and the Weierstrass theorem gives a minimizer of f on S . Every point outside S has value larger than $f(\mathbf{x}_0)$, so this minimizer is global. Strong convexity implies strict convexity, and a strictly convex function cannot have two distinct global minimizers. Thus the minimizer is unique. $\square$

Proposition 10.9 (Hessian characterization). *Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open and convex and let $f \in C^2(\mathcal{D})$. Then f is simultaneously μ -strongly convex and L -smooth on $\mathcal{D}$ if and only if*

$$\mu \mathbf{I} \preceq \nabla^2 f(\mathbf{x}) \preceq L \mathbf{I} \quad \text{for every } \mathbf{x} \in \mathcal{D}. \quad (10.17)$$

Proof. Suppose first that f is μ -strongly convex and L -smooth. Fix $\mathbf{x} \in \mathcal{D}$ and $\mathbf{v} \in \mathbb{R}^n$. Because $\mathcal{D}$ is open, $\mathbf{x} + t\mathbf{v} \in \mathcal{D}$ for all sufficiently small t . Applying strong convexity at $\mathbf{x}$ gives

$$f(\mathbf{x} + t\mathbf{v}) - f(\mathbf{x}) - t \nabla f(\mathbf{x})^\top \mathbf{v} \geq \frac{\mu t^2}{2} \|\mathbf{v}\|^2.$$

Divide by $t^2/2$ and let $t \rightarrow 0$. Taylor's theorem yields

$$\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} \geq \mu \|\mathbf{v}\|^2.$$

Smoothness gives

$$\frac{\|\nabla f(\mathbf{x} + t\mathbf{v}) - \nabla f(\mathbf{x})\|}{|t|} \leq L \|\mathbf{v}\|.$$

Letting $t \rightarrow 0$ shows $\|\nabla^2 f(\mathbf{x}) \mathbf{v}\| \leq L \|\mathbf{v}\|$. Since the Hessian is symmetric, all its eigenvalues are at most L ; together with the lower bound, this proves Equation (10.17).

Conversely, assume Equation (10.17). For $\mathbf{d} = \mathbf{y} - \mathbf{x}$, Taylor's formula with integral remainder gives

$$f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{d} = \int_0^1 (1-t) \mathbf{d}^\top \nabla^2 f(\mathbf{x} + t\mathbf{d}) \mathbf{d} dt \geq \frac{\mu}{2} \|\mathbf{d}\|^2.$$

Thus f is μ -strongly convex. Also,

$$\nabla f(\mathbf{y}) - \nabla f(\mathbf{x}) = \int_0^1 \nabla^2 f(\mathbf{x} + t\mathbf{d}) \mathbf{d} dt.$$

The upper matrix bound implies $\|\nabla^2 f(\mathbf{z})\|_{\text{op}} \leq L$ throughout the segment, so

$$\|\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})\| \leq \int_0^1 L \|\mathbf{d}\| dt = L \|\mathbf{y} - \mathbf{x}\|.$$

Hence f is L -smooth. □

For the nonquadratic slice ϕ in Equation (10.6), $\mu = 1/5$ and $L = 9/5$. Writing $f = \phi$ in the model notation, its lower and upper quadratic models therefore trap the function at every base point:

$$m_{\mu, \mathbf{x}}(\mathbf{d}) \leq f(\mathbf{x} + \mathbf{d}) \leq M_{L, \mathbf{x}}(\mathbf{d}),$$

where

$$m_{\mu, \mathbf{x}}(\mathbf{d}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{d} + \frac{\mu}{2} \|\mathbf{d}\|^2.$$

See an illustration in Figure 10.4.

Lemma 10.10 (Gradient and distance bounds for the optimality gap). *Let $f : \mathcal{D} \rightarrow \mathbb{R}$, where $\mathcal{D} = \mathbb{R}^n$, be μ -strongly convex and L -smooth, let $\mathbf{x}^*$ be its unique minimizer, and define $E(\mathbf{x}) := f(\mathbf{x}) - f(\mathbf{x}^*)$. Then*

$$2\mu E(\mathbf{x}) \leq \|\nabla f(\mathbf{x})\|^2 \leq 2LE(\mathbf{x}), \quad \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|^2 \leq E(\mathbf{x}) \leq \frac{L}{2} \|\mathbf{x} - \mathbf{x}^*\|^2. \quad (10.18)$$

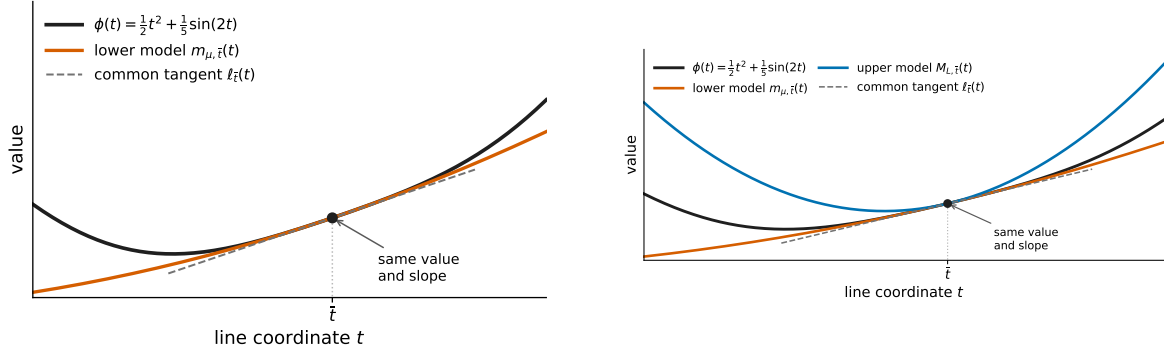


Figure 10.4: Strong convexity gives a quadratic lower model (left), while strong convexity and smoothness together place the nonquadratic function between lower and upper models (right).

Proof. Since $\mathbf{x}^*$ is an unconstrained differentiable minimizer, $\nabla f(\mathbf{x}^*) = \mathbf{0}$. Applying strong convexity with $\mathbf{x} = \mathbf{x}^*$ and $\mathbf{y} = \mathbf{x}$ gives

$$E(\mathbf{x}) \geq \frac{\mu}{2} \|\mathbf{x} - \mathbf{x}^*\|^2.$$

Applying the descent lemma with $\mathbf{x} = \mathbf{x}^*$ and $\mathbf{y} = \mathbf{x}$ gives

$$E(\mathbf{x}) \leq \frac{L}{2} \|\mathbf{x} - \mathbf{x}^*\|^2.$$

Next, minimize both sides of the strong-convexity inequality over $\mathbf{y} \in \mathbb{R}^n$. The lower quadratic model is minimized at $\mathbf{y} = \mathbf{x} - \nabla f(\mathbf{x})/\mu$, and hence

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) - \frac{1}{2\mu} \|\nabla f(\mathbf{x})\|^2.$$

Equivalently, $2\mu E(\mathbf{x}) \leq \|\nabla f(\mathbf{x})\|^2$.

Finally, take $\mathbf{y} = \mathbf{x} - \nabla f(\mathbf{x})/L$ in the descent lemma. Then

$$f(\mathbf{x}^*) \leq f(\mathbf{y}) \leq f(\mathbf{x}) - \frac{1}{2L} \|\nabla f(\mathbf{x})\|^2,$$

which is equivalent to $\|\nabla f(\mathbf{x})\|^2 \leq 2LE(\mathbf{x})$. □

Theorem 10.11 (Function-value linear convergence). *Let $f : \mathcal{D} \rightarrow \mathbb{R}$, where $\mathcal{D} = \mathbb{R}^n$, be μ -strongly convex and L -smooth, and let*

$$E^{(k)} := f(\mathbf{x}^{(k)}) - f(\mathbf{x}^*), \quad \rho := 1 - \frac{\mu}{L} \in [0, 1).$$

For either of the following choices,

$$E^{(k+1)} \leq \rho E^{(k)}, \quad E^{(k)} \leq \rho^k E^{(0)} :$$

(i) steepest descent with the fixed step $\alpha^{(k)} = 1/L$;

(ii) steepest descent with exact line search.

Thus the function-value residual converges Q -linearly unless the method has already reached the minimizer; when $\rho = 0$, it reaches the minimizer in at most one update.

Proof. Combining the strong-convexity lower model with the smoothness upper model for any two distinct points shows $\mu \leq L$, so $\rho \in [0, 1)$.

For the fixed step $1/L$, Equation (10.9) and Equation (10.18) give

$$E^{(k+1)} \leq E^{(k)} - \frac{1}{2L} \|\mathbf{g}^{(k)}\|^2 \leq \left(1 - \frac{\mu}{L}\right) E^{(k)}.$$

Induction yields $E^{(k)} \leq \rho^k E^{(0)}$.

At a nonoptimal iterate, strong convexity also gives, for $\alpha \geq 0$,

$$f(\mathbf{x}^{(k)} - \alpha \mathbf{g}^{(k)}) \geq f(\mathbf{x}^{(k)}) - \alpha \|\mathbf{g}^{(k)}\|^2 + \frac{\mu \alpha^2}{2} \|\mathbf{g}^{(k)}\|^2.$$

The right-hand side tends to $+\infty$ as $\alpha \rightarrow \infty$. The continuous line objective is therefore coercive on $[0, \infty)$ and attains an exact line-search minimum. Exact line search performs at least as well as the admissible trial step $1/L$ on the same negative-gradient ray. Its next function value is consequently no larger than the fixed-step value just analyzed, so the same one-step inequality and induction apply. $\square$

Corollary 10.12 (Geometric bounds for the iterates and gradients). *Under the assumptions of Theorem 10.11,*

$$\|\mathbf{x}^{(k)} - \mathbf{x}^*\| \leq \sqrt{\frac{2E^{(0)}}{\mu}} \rho^{k/2}, \quad \|\mathbf{g}^{(k)}\| \leq \sqrt{2LE^{(0)}} \rho^{k/2}.$$

Hence both the iterates and their gradients have geometric upper bounds.

Proof. The lower distance bound and the upper gradient bound in Equation (10.18), followed by $E^{(k)} \leq \rho^k E^{(0)}$, give the two displayed inequalities. $\square$

10.5.1 Condition number and landscape geometry

For a function that is strongly convex and smooth on $\mathbb{R}^n$, define the tight constants

$$\mu_f := \sup\{\mu > 0 : f \text{ is } \mu\text{-strongly convex}\}, \quad L_f := \inf\{L > 0 : f \text{ is } L\text{-smooth}\}.$$

When $0 < \mu_f \leq L_f < \infty$, its condition number is

$$\kappa_f := \frac{L_f}{\mu_f}. \tag{10.19}$$

Using non-tight valid constants still gives a correct convergence bound, but not the best one supplied by Theorem 10.11. With the tight constants, its factor is $1 - 1/\kappa_f$.

Corollary 10.13 (Iteration complexity from the condition number). *Use the tight constants in Theorem 10.11, and let $\varepsilon \in (0, 1)$. If $\kappa_f = 1$, the method reaches the minimizer in at most one update. If $\kappa_f > 1$, then*

$$k \geq \frac{\log(\varepsilon^{-1})}{\log((1 - 1/\kappa_f)^{-1})}$$

guarantees $E^{(k)} \leq \varepsilon E^{(0)}$. In particular, the simpler sufficient bound

$$k \geq \kappa_f \log(\varepsilon^{-1})$$

gives the complexity $O(\kappa_f \log(\varepsilon^{-1}))$.

Proof. If $E^{(0)} = 0$, the initial point is already the unique minimizer and the claim is immediate. Suppose $E^{(0)} > 0$. For $\kappa_f > 1$, [Theorem 10.11](#) gives

$$\frac{E^{(k)}}{E^{(0)}} \leq \left(1 - \frac{1}{\kappa_f}\right)^k.$$

Taking logarithms proves the first bound. The elementary inequality $1 - t \leq e^{-t}$, applied with $t = 1/\kappa_f$, gives

$$\left(1 - \frac{1}{\kappa_f}\right)^k \leq e^{-k/\kappa_f}.$$

The second displayed condition therefore also guarantees a relative gap at most ε . The case $\kappa_f = 1$ follows from the $\rho = 0$ case of [Theorem 10.11](#). $\square$

For the quadratic in [Equation \(10.11\)](#),

$$L_q = \lambda_{\max}(\mathbf{Q}), \quad \mu_q = \lambda_{\min}(\mathbf{Q}), \quad \kappa_q = \kappa(\mathbf{Q}) := \frac{\lambda_{\max}(\mathbf{Q})}{\lambda_{\min}(\mathbf{Q})}.$$

Indeed, these are the largest and smallest eigenvalues of its constant Hessian. The level set $q(\mathbf{x}) - q(\mathbf{x}^*) = c$ has principal semiaxis lengths $\sqrt{2c/\lambda_i(\mathbf{Q})}$. Consequently, the ratio of the longest to the shortest semiaxis is $\sqrt{\kappa(\mathbf{Q})}$. A large condition number is therefore both an algebraic measure of unequal curvature and a geometric measure of a flat, elongated landscape; see [Figure 10.5](#).

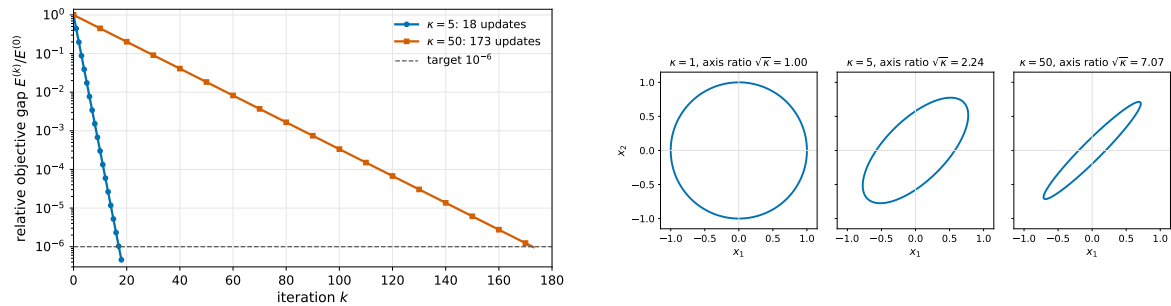


Figure 10.5: A larger condition number gives a slower worst-case gap factor (left) and a more elongated quadratic landscape (right).

Near a nondegenerate minimizer of a general twice continuously differentiable function, the Hessian gives the leading local quadratic landscape. Newton's method compensates for this anisotropy, whereas steepest descent does not; this explains why second-order information can determine local speed.

10.5.2 Sharper rate for positive-definite quadratics

The general factor $1 - 1/\kappa_f$ is convenient, but exact line search has a sharper bound on a positive-definite quadratic. Define

$$E^{(k)} := q(\mathbf{x}^{(k)}) - q(\mathbf{x}^*).$$

Since $\mathbf{Q}(\mathbf{x}^{(k)} - \mathbf{x}^*) = \mathbf{g}^{(k)}$,

$$E^{(k)} = \frac{1}{2}(\mathbf{x}^{(k)} - \mathbf{x}^*)^\top \mathbf{Q}(\mathbf{x}^{(k)} - \mathbf{x}^*) = \frac{1}{2}(\mathbf{g}^{(k)})^\top \mathbf{Q}^{-1} \mathbf{g}^{(k)}. \quad (10.20)$$

Lemma 10.14 (Kantorovich inequality). *Let $\mathbf{Q} \succ 0$, let $\lambda_{\min}$ and $\lambda_{\max}$ be its extreme eigenvalues, and let $\mathbf{z} \neq \mathbf{0}$. Then*

$$\frac{(\mathbf{z}^\top \mathbf{z})^2}{(\mathbf{z}^\top \mathbf{Q} \mathbf{z})(\mathbf{z}^\top \mathbf{Q}^{-1} \mathbf{z})} \geq \frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min} + \lambda_{\max})^2}.$$

Proof. By the spectral theorem, write $\mathbf{Q} = \mathbf{U} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{U}^\top$, where $\mathbf{U}$ is orthogonal and $\lambda_{\min} \leq \lambda_i \leq \lambda_{\max}$. Let $\mathbf{y} = \mathbf{U}^\top \mathbf{z}$ and

$$w_i := \frac{y_i^2}{\mathbf{z}^\top \mathbf{z}}, \quad a := \sum_i w_i \lambda_i, \quad b := \sum_i \frac{w_i}{\lambda_i}.$$

Then $w_i \geq 0$, $\sum_i w_i = 1$, and the ratio on the left is $1/(ab)$. For every $\lambda \in [\lambda_{\min}, \lambda_{\max}]$,

$$(\lambda - \lambda_{\min})(\lambda - \lambda_{\max}) \leq 0,$$

or equivalently,

$$\lambda + \frac{\lambda_{\min}\lambda_{\max}}{\lambda} \leq \lambda_{\min} + \lambda_{\max}.$$

Multiplying by w_i and summing gives $a + \lambda_{\min}\lambda_{\max}b \leq \lambda_{\min} + \lambda_{\max}$. Since $(a - \lambda_{\min}\lambda_{\max}b)^2 \geq 0$,

$$4\lambda_{\min}\lambda_{\max}ab \leq (a + \lambda_{\min}\lambda_{\max}b)^2 \leq (\lambda_{\min} + \lambda_{\max})^2.$$

Taking reciprocals proves the result. $\square$

Theorem 10.15 (Exact-search quadratic rate). *Apply exact-line-search steepest descent to Equation (10.11). At every iteration for which $\mathbf{g}^{(k)} \neq \mathbf{0}$,*

$$\frac{E^{(k+1)}}{E^{(k)}} = 1 - \frac{((\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)})^2}{((\mathbf{g}^{(k)})^\top \mathbf{Q} \mathbf{g}^{(k)})(\mathbf{g}^{(k)})^\top \mathbf{Q}^{-1} \mathbf{g}^{(k)}} \leq \left(\frac{\kappa(\mathbf{Q}) - 1}{\kappa(\mathbf{Q}) + 1} \right)^2. \quad (10.21)$$

Proof. Using $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha^{(k)} \mathbf{g}^{(k)}$, expand

$$\begin{aligned} E^{(k+1)} &= E^{(k)} - \alpha^{(k)} (\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)} + \frac{(\alpha^{(k)})^2}{2} (\mathbf{g}^{(k)})^\top \mathbf{Q} \mathbf{g}^{(k)} \\ &= E^{(k)} - \frac{1}{2} \frac{((\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)})^2}{(\mathbf{g}^{(k)})^\top \mathbf{Q} \mathbf{g}^{(k)}}, \end{aligned}$$

where the second equality uses Equation (10.12). Divide by $E^{(k)} = \frac{1}{2}(\mathbf{g}^{(k)})^\top \mathbf{Q}^{-1} \mathbf{g}^{(k)} > 0$ to obtain the equality in Equation (10.21). The Kantorovich inequality gives the bound, because

$$1 - \frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min} + \lambda_{\max})^2} = \left(\frac{\kappa(\mathbf{Q}) - 1}{\kappa(\mathbf{Q}) + 1} \right)^2.$$

□

For $r > 1$, the family in Equation (10.13) attains this bound at every iteration. Put $s := (r - 1)/(r + 1)$ and use the eigen-coordinates $\mathbf{y}^{(k)} := \mathbf{U}^\top \mathbf{x}^{(k)}$. We claim that

$$\alpha^{(k)} = \frac{2}{r + 1}, \quad \mathbf{y}^{(k)} = s^k (4, 4(-1)^k/r)^\top. \quad (10.22)$$

The formula for $\mathbf{y}^{(0)}$ is Equation (10.14). If the iterate formula holds at k , then its gradient in eigen-coordinates is

$$\begin{pmatrix} 1 & 0 \\ 0 & r \end{pmatrix} \mathbf{y}^{(k)} = 4s^k (1, (-1)^k)^\top.$$

Consequently,

$$\alpha^{(k)} = \frac{32s^{2k}}{16(1 + r)s^{2k}} = \frac{2}{1 + r}.$$

The two eigen-components of the next iterate are multiplied by

$$1 - \frac{2}{1 + r} = s, \quad 1 - \frac{2r}{1 + r} = -s,$$

respectively. This proves the iterate formula at $k + 1$, and hence Equation (10.22) by induction. Consequently,

$$\frac{q_r(\mathbf{x}^{(k+1)})}{q_r(\mathbf{x}^{(k)})} = s^2 = \left(\frac{r - 1}{r + 1} \right)^2.$$

Since $\kappa(\mathbf{Q}_r) = r$, equality holds in Equation (10.21). Hence the factor cannot be improved using only the extreme eigenvalues. This result is sharper than the general strongly-convex smooth factor because it also exploits the exact quadratic structure and exact line search.

The relative objective gap after k updates is s^{2k} . Thus the first integer k for which it is strictly below 10^{-6} is

$$\left\lceil \frac{\log(10^{-6})}{2 \log s} \right\rceil + 1,$$

which equals 18 for $r = 5$ and 173 for $r = 50$. When $r = 1$, the quadratic is isotropic and steepest descent reaches its minimizer in one update, so no nontrivial asymptotic recurrence remains.

10.6 Preconditioned steepest descent

The Euclidean method treats every coordinate direction equally. A fixed symmetric positive-definite matrix $\mathbf{B}$ supplies a different geometry without requiring a new line-search framework. At $\mathbf{x}^{(k)}$, consider

$$m_{\mathbf{B}}^{(k)}(\mathbf{d}) := f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{B} \mathbf{d}. \quad (10.23)$$

This strictly convex model has the unique minimizer determined by

$$\mathbf{B}\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}, \quad \mathbf{d}^{(k)} = -\mathbf{B}^{-1}\mathbf{g}^{(k)}. \quad (10.24)$$

The matrix should be applied by solving the linear system; its inverse need not be formed explicitly.

Proposition 10.16 (Descent and steepest decrease in the $\mathbf{B}$ -norm). *Let $\mathbf{B} \succ 0$, define $\|\mathbf{u}\|_{\mathbf{B}} := (\mathbf{u}^\top \mathbf{B} \mathbf{u})^{1/2}$, and suppose $\mathbf{g}^{(k)} \neq \mathbf{0}$. The direction in Equation (10.24) is a strict descent direction. Moreover,*

$$\min_{\|\mathbf{u}\|_{\mathbf{B}}=1} (\mathbf{g}^{(k)})^\top \mathbf{u} = -\sqrt{(\mathbf{g}^{(k)})^\top \mathbf{B}^{-1} \mathbf{g}^{(k)}},$$

and the unique minimizing unit direction is

$$\mathbf{u}_* = -\frac{\mathbf{B}^{-1}\mathbf{g}^{(k)}}{\sqrt{(\mathbf{g}^{(k)})^\top \mathbf{B}^{-1} \mathbf{g}^{(k)}}}.$$

Proof. Because $\mathbf{B}\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$,

$$(\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)} = -(\mathbf{d}^{(k)})^\top \mathbf{B}\mathbf{d}^{(k)} < 0;$$

the inequality is strict because $\mathbf{g}^{(k)} \neq \mathbf{0}$ implies $\mathbf{d}^{(k)} \neq \mathbf{0}$. For any $\mathbf{B}$ -unit vector $\mathbf{u}$, Cauchy–Schwarz in the inner product $\langle \mathbf{v}, \mathbf{u} \rangle_{\mathbf{B}} = \mathbf{v}^\top \mathbf{B} \mathbf{u}$ gives

$$\begin{aligned} (\mathbf{g}^{(k)})^\top \mathbf{u} &= (\mathbf{B}^{-1}\mathbf{g}^{(k)})^\top \mathbf{B} \mathbf{u} \\ &\geq -\sqrt{(\mathbf{g}^{(k)})^\top \mathbf{B}^{-1} \mathbf{g}^{(k)}} \|\mathbf{u}\|_{\mathbf{B}}. \end{aligned}$$

Equality holds precisely at the displayed $\mathbf{u}_*$. This proves both claims. $\square$

Because the direction is descending, exact or Armijo line search can be used without changing the algorithms of Equations (10.2) and (10.3). The choice $\mathbf{B} = \mathbf{I}$ recovers ordinary steepest descent. The point of a nontrivial $\mathbf{B}$ is to make the landscape closer to isotropic in the geometry seen by the algorithm.

10.6.1 Effective conditioning on a quadratic

Apply the method to Equation (10.11) and let $\mathbf{e}^{(k)} := \mathbf{x}^{(k)} - \mathbf{x}^*$. With the symmetric positive-definite square root of $\mathbf{B}$, define

$$\mathbf{y}^{(k)} := \mathbf{B}^{1/2} \mathbf{e}^{(k)}, \quad \tilde{\mathbf{Q}} := \mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}. \quad (10.25)$$

The transformed matrix $\tilde{\mathbf{Q}}$ is symmetric positive definite.

Theorem 10.17 (Exact-search rate after preconditioning). *Let $\mathbf{B} \succ 0$ be fixed. Apply $\mathbf{d}^{(k)} = -\mathbf{B}^{-1}\mathbf{g}^{(k)}$ with exact line search to the quadratic in Equation (10.11). At every nonoptimal iterate,*

$$\frac{E^{(k+1)}}{E^{(k)}} \leq \left(\frac{\kappa(\tilde{\mathbf{Q}}) - 1}{\kappa(\tilde{\mathbf{Q}}) + 1} \right)^2.$$

Thus $\kappa(\tilde{Q})$, rather than $\kappa(Q)$, is the effective condition number.

Proof. Since $\mathbf{g}^{(k)} = Q\mathbf{e}^{(k)}$, the preconditioned update gives

$$\mathbf{e}^{(k+1)} = \mathbf{e}^{(k)} - \alpha^{(k)} B^{-1} Q \mathbf{e}^{(k)}.$$

Multiplying by $B^{1/2}$ and using $\mathbf{e}^{(k)} = B^{-1/2} \mathbf{y}^{(k)}$ yields

$$\mathbf{y}^{(k+1)} = \mathbf{y}^{(k)} - \alpha^{(k)} \tilde{Q} \mathbf{y}^{(k)}.$$

Furthermore,

$$E^{(k)} = \frac{1}{2} (\mathbf{e}^{(k)})^\top Q \mathbf{e}^{(k)} = \frac{1}{2} (\mathbf{y}^{(k)})^\top \tilde{Q} \mathbf{y}^{(k)}.$$

The map between $\mathbf{e}$ and $\mathbf{y}$ is invertible and preserves the line parameter and objective values. Hence exact line search in the original variables is ordinary exact-search steepest descent on the transformed quadratic. Applying [Theorem 10.15](#) with $\tilde{Q}$ proves the result. $\square$

Example 10.2 (Jacobi scaling on a rotated quadratic). Let

$$Q = U_\theta \begin{pmatrix} 1 & 0 \\ 0 & 50 \end{pmatrix} U_\theta^\top, \quad U_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \theta = 10^\circ,$$

and choose the Jacobi preconditioner $B_J = \text{diag}(Q)$. Direct eigenvalue computation gives

$$\kappa(Q) = 50, \quad \kappa(B_J^{-1/2} Q B_J^{-1/2}) \approx 7.48365.$$

An illustration of the corresponding coordinate transformation is given in [Figure 10.6](#). Start from the point satisfying $U_\theta^\top \mathbf{x}^{(0)} = (4, 4/50)^\top$, and use exact line search. To reduce the relative objective gap to at most 10^{-6} , ordinary SD uses 173 updates, Jacobi-preconditioned SD uses 19, and the ideal choice $B = Q$ uses one; see [Figure 10.7](#). This comparison measures iterations only; the cost of solving with B must still be counted.

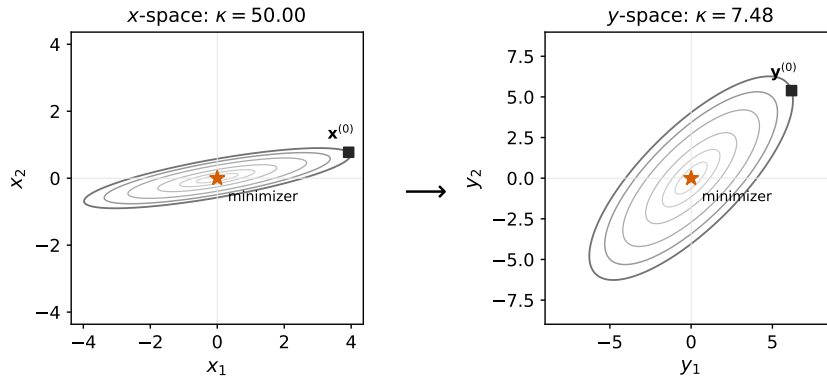


Figure 10.6: The coordinate transformation makes the same quadratic more isotropic. Ordinary SD in the transformed coordinates is preconditioned SD in the original coordinates.

Two limiting choices clarify the idea:

- $B = I$ gives $\tilde{Q} = Q$, so nothing changes.

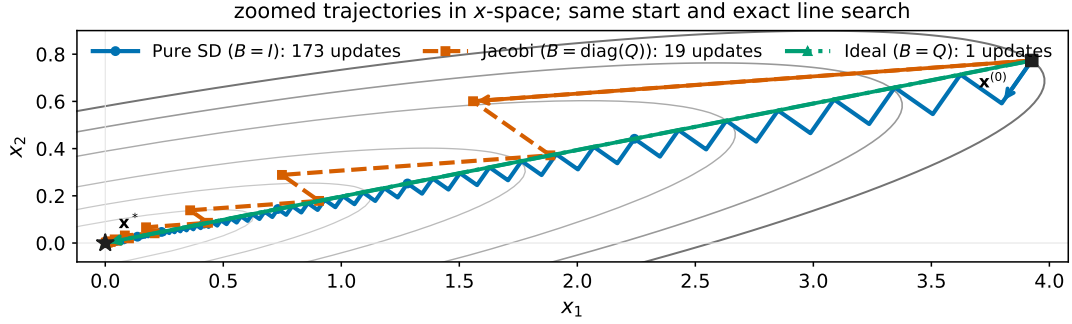


Figure 10.7: Exact-search trajectories for the example in [Example 10.2](#). The preconditioner changes the search ray, while line search selects a distance along that ray.

- $B = Q$ gives $\tilde{Q} = I$ and $d^{(k)} = -Q^{-1}g^{(k)} = x^* - x^{(k)}$. The next exact-search iterate is the minimizer; this is the Newton step.

Between these extremes, one seeks a matrix that is inexpensive to store and solve with but still approximates the important curvature. Common choices include the diagonal or Jacobi matrix $\text{diag}(Q)$, block diagonal matrices, preconditioners built from incomplete Cholesky factors, and other cheap positive-definite approximations. A richer approximation may reduce the effective condition number, but its construction, storage, and linear solves also cost more. Preconditioning is worthwhile only when the reduction in iterations compensates for that extra work.

Warning

Diagonal scaling need not help when the poor conditioning is caused by a rotation rather than unequal coordinate scales. For the 45° -rotated matrix Q_r in [Equation \(10.13\)](#),

$$\text{diag}(Q_r) = \frac{1+r}{2}I.$$

The Jacobi direction is therefore only a positive scalar multiple of the ordinary negative gradient. Exact line search absorbs that scalar, so the iterates and the convergence rate are unchanged.

10.7 A smooth convex example with sublinear convergence

Strong convexity is sufficient for the linear rate above, but smooth convexity alone is not. Consider

$$\varphi(x) := e^{-x^2} + x^2 - 1, \quad x \in \mathcal{D} = \mathbb{R}. \quad (10.26)$$

Its first two derivatives are

$$\varphi'(x) = 2x(1 - e^{-x^2}), \quad \varphi''(x) = 2 + (4x^2 - 2)e^{-x^2}.$$

Writing $s = x^2$, define

$$\psi(s) := 2 + (4s - 2)e^{-s}.$$

Then $\psi(0) = 0$, $\psi'(s) = (6 - 4s)e^{-s}$, so ψ increases on $[0, 3/2]$ and then decreases to its limit 2. Hence

$$0 \leq \varphi''(x) \leq L, \quad L := 2 + 4e^{-3/2}.$$

Thus φ is convex and L -smooth. Since $\varphi''(0) = 0$, it is not strongly convex. Its unique minimizer is $x^* = 0$, because $\varphi'(x)$ has the sign of x .

Apply steepest descent with $\alpha^{(k)} = 1/L$ from any $x^{(0)} > 0$. Set $c := 2/L \in (0, 1)$. Then

$$x^{(k+1)} = x^{(k)} [1 - c(1 - e^{-(x^{(k)})^2})]. \quad (10.27)$$

The bracket lies strictly between $1 - c$ and 1. Therefore $x^{(k)}$ is positive and decreasing. If its limit were $\ell > 0$, passing to the limit in Equation (10.27) would give $c\ell(1 - e^{-\ell^2}) = 0$, a contradiction. Hence $x^{(k)} \downarrow 0$.

Taylor expansion at zero gives

$$1 - e^{-x^2} = x^2 + O(x^4), \quad \varphi(x) = \frac{x^4}{2} + O(x^6).$$

In particular,

$$\frac{x^{(k+1)}}{x^{(k)}} = 1 - c(x^{(k)})^2 + O((x^{(k)})^4).$$

Using $(1 - u)^{-2} = 1 + 2u + O(u^2)$, it follows that

$$\lim_{k \rightarrow \infty} \left(\frac{1}{(x^{(k+1)})^2} - \frac{1}{(x^{(k)})^2} \right) = 2c = \frac{4}{L}.$$

To make the order statement explicit, the positive sequence inside the limit is bounded above and below by positive constants for all sufficiently large k . Summing those bounds gives

$$\frac{1}{(x^{(k)})^2} = \Theta(k), \quad (x^{(k)})^2 = \Theta(k^{-1}).$$

For $E^{(k)} := \varphi(x^{(k)}) - \varphi(0)$, the second Taylor expansion therefore yields

$$E^{(k)} = \Theta(k^{-2}), \quad \lim_{k \rightarrow \infty} \frac{E^{(k+1)}}{E^{(k)}} = 1. \quad (10.28)$$

The ratio limit follows because $x^{(k+1)}/x^{(k)} \rightarrow 1$ and $\varphi(x)/x^4 \rightarrow 1/2$. Thus the function values converge Q -sublinearly, by which we mean here that $E^{(k)} \rightarrow 0$ but $E^{(k+1)}/E^{(k)} \rightarrow 1$.

This example also identifies the exact step in the linear-rate proof that fails. As $x \rightarrow 0$,

$$\frac{|\varphi'(x)|^2}{2(\varphi(x) - \varphi(0))} \sim 4x^2 \rightarrow 0.$$

Therefore no uniform $\mu > 0$ can make $\|\nabla\varphi(x)\|^2 \geq 2\mu(\varphi(x) - \varphi(0))$ near the minimizer. The descent inequality remains valid, but the gradient-to-gap bound needed for a fixed linear contraction is absent.

A numerical run from $x^{(0)} = 1$, with $\alpha^{(k)} = 1/L$, illustrates the ratio approaching one:

k	$x^{(k)}$	$E^{(k)}$	$E^{(k)}/E^{(k-1)}$
10	0.245882	1.7913×10^{-3}	0.835862
100	0.083870	2.4682×10^{-5}	0.980589
1000	0.026842	2.5948×10^{-7}	0.998008
10000	0.008502	2.6121×10^{-9}	0.999800

See also Figure 10.8.

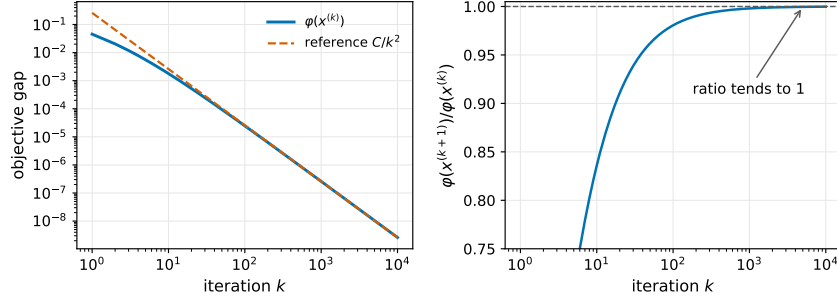


Figure 10.8: Fixed-step SD on the smooth convex function $\varphi(x) = e^{-x^2} + x^2 - 1$. The objective gap decays as $\Theta(k^{-2})$, while the one-step ratio approaches 1.

Warning

Strict convexity does not imply strong convexity. A unique minimizer and successful numerical convergence do not supply a positive uniform curvature lower bound. The assumptions of a linear-rate theorem must be verified before its factor is used.

10.8 From full gradients to stochastic gradients

Large machine-learning objectives often have finite-sum form

$$F(\mathbf{w}) := \frac{1}{N} \sum_{i=1}^N \ell_i(\mathbf{w}), \quad \mathbf{w} \in \mathcal{D}. \quad (10.29)$$

Computing ∇F uses all N data terms. A mini-batch method instead samples indices $I_1^{(k)}, \dots, I_b^{(k)}$, conditionally independently and uniformly from $\{1, \dots, N\}$ after $\mathbf{w}^{(k)}$ is fixed. Denote the resulting multiset by $\mathcal{B}^{(k)}$. The mini-batch gradient is

$$\hat{\mathbf{g}}^{(k)} := \frac{1}{|\mathcal{B}^{(k)}|} \sum_{i \in \mathcal{B}^{(k)}} \nabla \ell_i(\mathbf{w}^{(k)}).$$

Proposition 10.18 (Unbiased mini-batch gradient). *Conditionally on $\mathbf{w}^{(k)}$,*

$$\mathbb{E}[\hat{\mathbf{g}}^{(k)} \mid \mathbf{w}^{(k)}] = \nabla F(\mathbf{w}^{(k)}).$$

Moreover, if

$$V(\mathbf{w}) := \frac{1}{N} \sum_{i=1}^N \|\nabla \ell_i(\mathbf{w}) - \nabla F(\mathbf{w})\|^2,$$

then sampling with replacement gives

$$\mathbb{E} \left[\left\| \hat{\mathbf{g}}^{(k)} - \nabla F(\mathbf{w}^{(k)}) \right\|^2 \mid \mathbf{w}^{(k)} \right] = \frac{V(\mathbf{w}^{(k)})}{b}.$$

Proof. Condition on $\mathbf{w}^{(k)}$ and abbreviate it by $\mathbf{w}$. Uniform sampling gives

$$\mathbb{E}[\nabla \ell_{I_j^{(k)}}(\mathbf{w})] = \frac{1}{N} \sum_{i=1}^N \nabla \ell_i(\mathbf{w}) = \nabla F(\mathbf{w}).$$

Linearity of expectation proves unbiasedness. Let $\mathbf{z}_j := \nabla \ell_{I_j^{(k)}}(\mathbf{w}) - \nabla F(\mathbf{w})$. The vectors $\mathbf{z}_j$ are independent, have mean zero, and satisfy $\mathbb{E} \|\mathbf{z}_j\|^2 = V(\mathbf{w})$. Therefore the cross terms have zero expectation, and

$$\mathbb{E} \left\| b^{-1} \sum_{j=1}^b \mathbf{z}_j \right\|^2 = \frac{1}{b^2} \sum_{j=1}^b \mathbb{E} \|\mathbf{z}_j\|^2 = \frac{V(\mathbf{w})}{b}.$$

□

Stochastic gradient descent uses

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - \eta^{(k)} \hat{\mathbf{g}}^{(k)}. \quad (10.30)$$

A larger batch costs more per update but reduces the gradient noise. One *epoch* is roughly one pass through all N samples. Learning rates are commonly held nearly constant at first and then reduced; their scale and schedule are central practical choices.

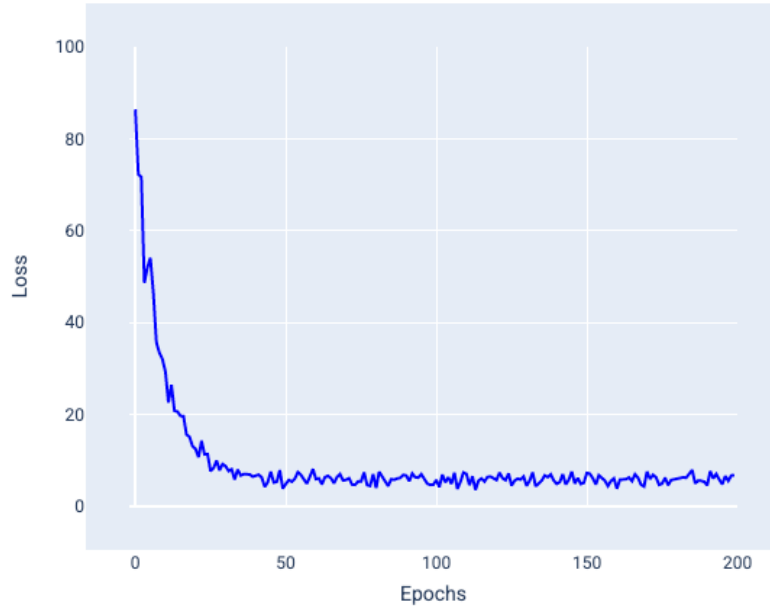


Figure 10.9: A representative neural-network training curve. Random mini-batches make the observed loss fluctuate even while its overall trend decreases. Source: [Google Machine Learning Crash Course, Figure 24](#) (CC BY 4.0).

Warning

The mini-batch vector is an unbiased estimate of the full gradient, but it need not be a descent direction for F at every update. Evaluating F repeatedly inside a deterministic line search would also remove much of the computational advantage. Consequently, practical stochastic methods usually use prescribed learning-rate schedules rather than exact or Armijo line search.

Momentum averages current and past stochastic gradients before updating. AdaGrad accumulates coordinatewise squared gradients, whereas RMSProp uses an exponentially weighted

average so very old gradients gradually lose influence. Adam combines a momentum-like first moment with an RMSProp-like second moment and adds bias correction. These variants can behave very differently in deep learning, but they preserve the basic pattern: sample a mini-batch, form a gradient estimate, and take an inexpensive update.

Chapter summary

- Replacing the Hessian in the Newton model by $\mathbf{I}$ gives the negative-gradient ray. Exact line search absorbs a positive rescaling; Armijo backtracking does so only when its trial grid is rescaled as well.
- The negative gradient is the Euclidean direction of steepest local decrease and is a strict descent direction whenever the gradient is nonzero.
- Exact line search minimizes on the current ray; Armijo backtracking seeks sufficient decrease. For an L -smooth objective, $\alpha \leq 1/L$ satisfies Armijo with $\sigma = 1/2$.
- Exact line search makes successive gradients orthogonal. On flat quadratic landscapes this creates zig-zagging, while Newton solves the quadratic in one full step.
- For a μ -strongly convex, L -smooth objective, fixed step $1/L$ and exact-search steepest descent have function-value factor at most $1 - \mu/L$. The tight ratio $\kappa_f = L_f/\mu_f$ measures conditioning.
- A fixed $\mathbf{B} \succ 0$ changes the steepest direction to $-\mathbf{B}^{-1}\mathbf{g}^{(k)}$. On a quadratic, its speed is controlled by the effective condition number of $\mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2}$, balanced against the cost of solving with $\mathbf{B}$.
- Smooth convexity alone does not guarantee a linear rate. The non-strongly-convex example above has Q -sublinear objective-gap decay.
- Stochastic gradient descent replaces the full gradient by a cheap, noisy mini-batch estimate and normally uses a prescribed learning rate.

10.9 Exercises

Exercise 10.1 (Unit steepest direction). Let

$$f(x_1, x_2) := (x_1 - 1)^2 + 2(x_1 + x_2)^2.$$

At $\mathbf{x} = (0, 1)^\top$, compute both the Euclidean unit direction of steepest decrease and the unnormalized steepest-descent search direction.

Exercise 10.2 (Scaling an isotropic model and its line search). Consider

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Set $\mathbf{g}^{(0)} = \nabla q(\mathbf{x}^{(0)})$. For $c > 0$, minimize

$$m_c^{(0)}(\mathbf{d}) = q(\mathbf{x}^{(0)}) + (\mathbf{g}^{(0)})^\top \mathbf{d} + \frac{c}{2} \|\mathbf{d}\|^2$$

to obtain a direction $\mathbf{d}_c^{(0)}$. Let $\alpha_c^{(0)}$ be the exact line-search step length along this direction. Prove that $\alpha_c^{(0)} = c\alpha_1^{(0)}$, and hence that the next exact-search iterate is independent of c . Explain why Armijo backtracking with the same numerical initial trial step need not have this invariance.

Exercise 10.3 (One exact and one Armijo search). Let

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \mathbf{x} - \begin{pmatrix} 1 \\ 0 \end{pmatrix}^\top \mathbf{x}, \quad \mathbf{x}^{(0)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Apply steepest descent, so $\mathbf{d}^{(0)} = -\nabla f(\mathbf{x}^{(0)})$. Compute $\mathbf{d}^{(0)}$, the exact line-search step length $\alpha^{(0)}$, and $\mathbf{x}^{(1)}$. Then apply Armijo backtracking with $\bar{\alpha} = 1$, $\beta = 1/2$, and $\sigma = 1/2$; list all tested steps and identify the first accepted one.

Exercise 10.4 (Strong convexity, smoothness, and a rate). Consider

$$f(x) = \log(\cosh x) + \frac{1}{2}x^2, \quad x \in \mathbb{R}.$$

Show that the tight constants are $\mu_f = 1$ and $L_f = 2$. Find κ_f , identify the minimizer, and give the function-value bound from [Theorem 10.11](#) for the fixed step $1/2$.

Exercise 10.5 (A weighted steepest direction). At an iterate, suppose

$$\mathbf{g}^{(k)} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Solve $\mathbf{B}\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$, verify directly that $\mathbf{d}^{(k)}$ is descending, and normalize it to obtain the $\mathbf{B}$ -unit direction of steepest decrease. Next normalize the Euclidean direction $-\mathbf{g}^{(k)}$ to have $\mathbf{B}$ -norm one. Compare the two directional derivatives and explain the difference.

Exercise 10.6 (Orientation and diagonal preconditioning). For

$$\mathbf{Q}_1 = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Q}_2 = \frac{1}{5} \begin{pmatrix} 37 & 16 \\ 16 & 13 \end{pmatrix},$$

consider $q_i(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{Q}_i \mathbf{x}$. Verify that the two matrices have the same eigenvalues, and compute their tight strong-convexity constants, tight smoothness constants, condition numbers, and level-set semiaxis ratios. For each i , now choose the Jacobi preconditioner $\mathbf{B}_i = \text{diag}(\mathbf{Q}_i)$. Compute $\tilde{\mathbf{Q}}_i = \mathbf{B}_i^{-1/2} \mathbf{Q}_i \mathbf{B}_i^{-1/2}$, its effective condition number, and the corresponding sharp exact-search factor. Explain why a coordinate-dependent preconditioner can behave differently on two quadratics with the same eigenvalues.

Exercise 10.7 (A different sublinear example, challenging). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) =$

$x^4/4$. Starting from $x^{(0)} \in (0, 1]$, apply

$$x^{(k+1)} = x^{(k)} - \frac{1}{3}f'(x^{(k)}).$$

Show that the iterates remain in $[0, 1]$, decrease to zero, and satisfy

$$(x^{(k)})^2 = \Theta(k^{-1}), \quad f(x^{(k)}) = \Theta(k^{-2}), \quad \frac{f(x^{(k+1)})}{f(x^{(k)})} \rightarrow 1.$$

Exercise 10.8 (Mini-batch mean and variance). Let

$$F(x) = \frac{1}{2}(\ell_1(x) + \ell_2(x)), \quad \ell_1(x) = \frac{1}{2}(x-1)^2, \quad \ell_2(x) = \frac{1}{2}(x+2)^2.$$

At $x = 0$, sample component gradients uniformly with replacement. For batch sizes $b = 1$ and $b = 2$, compute the mean and variance of the mini-batch estimator and compare its mean with $F'(0)$. Verify directly that the variance is divided by b .

Exercise 10.9 (Equality in the quadratic rate, advanced). Let

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{Q} \mathbf{x}, \quad \mathbf{Q} \succ 0,$$

where $\lambda_{\min}(\mathbf{Q}) < \lambda_{\max}(\mathbf{Q})$. At a nonoptimal iterate define $\mathbf{g}^{(k)} = \nabla q(\mathbf{x}^{(k)})$. Prove that one exact steepest-descent step attains equality in the upper bound of [Equation \(10.21\)](#) if and only if $\mathbf{g}^{(k)}$ has no component outside the two extreme eigenspaces and the Euclidean norms of its projections onto those eigenspaces are equal. Deduce the equality case $\mathbf{g}^{(k)} = a(\mathbf{u}_{\min} + \mathbf{u}_{\max})$, where $a \neq 0$ and $\mathbf{u}_{\min}, \mathbf{u}_{\max}$ are unit extreme-eigenvalue eigenvectors.